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Reflexive digital twins and behavioral governance in data-driven systems

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Article

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Abstract

Digital twins are increasingly used to model and optimise complex systems, yet their behavioural and institutional implications remain fragmented across disciplines. Here, we develop Reflexive Digital Twin Marketing Theory (RDTMT), conceptualising marketing as a socio-technical learning system that co-evolves with consumers, algorithms, and governance arrangements through continuous feedback. Using an Empirics-First approach, we conduct a PRISMA-ScR–guided scoping review across Web of Science, Scopus, PubMed, and Google Scholar (2010–2025), identifying 742 records and synthesising 55 studies at the intersection of healthcare digital twins, marketing analytics, and ethical AI governance. Co-occurrence mapping (VOS) reveals four convergent clusters—continuous synchronisation, adaptive intelligence, reflexive governance, and societal co-evolution—indicating that "twin-like" systems reshape behaviour by structuring how people are represented, predicted, and acted upon. Building on these regularities, RDTMT introduces three constructs—Digital Reflexivity, Ethical Feedback Elasticity and the Value Coherence Index—to explain when predictive personalisation aligns with stakeholder trust and social legitimacy. This framework reframes marketing not as unidirectional influence but as reflexive governance in data-rich environments, offering testable propositions for responsible behavioural prediction and feedback-driven institutional design.

Keywords:

Live marketing; Ethical AI; Stakeholder Co-Evolution; Marketing Analytics; Theoretical Abstraction; Value Coherence

Introduction:

The importance of reflexive intelligence systems, which couple data, algorithms, and human intelligence, has emerged in learning, appraisal, and adaptation through recursive processes. In marketing and health care, intelligence systems and networks have shifted from traditional linear approaches to dynamic environments in which knowledge building and management occur through shared human and algorithmic processes. One of the best examples of such processes is the Digital Twin, a virtual replica created through engineering processes specifically for the purpose of optimization.[1]. The Digital Twin is a prime example of an essential ingredient of "smart" health. It works as the "mirror of the patient's system," providing a real-time algorithmic simulation of the patient. This mirror, developed in a contemporary fashion, combines biological, behavioural, and environmental inputs to enable simulation and prediction. The current approach would redefine the 'patient' as the "status of treatment," developed within a care paradigm in which humans, institutions, and intelligent systems collaborate in the 'co-production of care.'[2] In such a paradigm, the patient would be seen as an 'active source of data' in the process of 'managing their digital identity' through feedback and consent.

Similarly, in the research ecosystem, the Empirics-first (EF)[3]can be seen as a Digital Twin of inquiry—a reflective, adaptive system that mirrors the methodological, ethical, and epistemic aspects of research. Like a Digital Twin (DT) that recalibrates a physiological model with real-time data, a Reflexive Evaluation Framework (REF) continuously adapts and improves by modelling the changing research dynamics—tracking outputs, learning paths, contextual relationships, and ethical considerations.[4]

These reframing shifts evaluation from a static, judgment-focused process to a dynamic epistemic ecosystem that can sense, model, and enhance its own functions. The REF thus serves

as both a reflection and a guide for research: it displays performance metrics while also steering ethical, methodological, and social responsiveness.[5]. In this way, it aligns with the main principles of the Digital Twin approach—such as simulation, feedback, transparency, and co-evolution—helping research remain rigorous, context-aware, stakeholder-accountable, and adaptable to its evolving complexities, as shown in Table 1.

Table 1: Conceptual Bridge: Digital Twin–Evaluation Framework Alignment

Continuous modelling of patient physiology	Continuous modelling of research quality and impact
Integration of multimodal data (biological, behavioural, environmental)	Integrating multiple sources of evidence, including quantitative metrics, qualitative insights, and stakeholder feedback.
Feedback loops for predictive adaptation	Reflexive loops for methodological and ethical recalibration
Patient as co-producer of care	Researcher and stakeholder as co-producers of knowledge
Adaptive governance ensuring trust and accountability	Adaptive evaluation ensuring transparency, inclusion, and ethical legitimacy

Essentially, the Digital Twin serves as both a technological analogy and a conceptual model for a new type of Evaluation Framework—one that learns, adapts, and corrects itself in real time. [6]A Reflexive Evaluation Framework (REF) captures the essence of the Digital Twin: it is predictive (capable of forecasting outcomes), participatory (including stakeholder input), and flexible (allowing ethical adjustments).[7]. In this setup, evaluation shifts from a backward-looking audit to a dynamic, living epistemic structure—an intelligent twin that co-evolves with the systems it aims to understand and enhance.

Conceptual Foundations: The Digital Twin as a Reflexive Evaluation Framework (REF)

From Static Assessment to Reflexive Intelligence

The 21st century signifies a pivotal change in how organizations, researchers, and policymakers approach evaluation, learning, and adaptation. Historically, evaluation frameworks served as linear, retrospective tools—used to measure research outcomes after completion. They prioritized procedural aspects, including compliance, methodological rigor, and performance metrics.[8] However, this traditional approach is becoming less effective in a time marked by complex, data-rich, and constantly changing systems.

The rapid integration of data infrastructures, machine learning, and algorithmic reasoning has led to what can be called reflexive intelligence systems—ecosystems that learn about themselves while they act. In these environments, evaluation cannot remain external to the system it assesses; it must become adaptive, participatory, and co-evolving.[9]. This gives rise to what we might call a Reflexive Evaluation Framework (REF)—a dynamic, living system that incorporates feedback loops, ethical reasoning, and real-time data synthesis to model and guide the ongoing development of research.

Within this broader epistemic transformation, the **Digital Twin (DT)**—a real-time digital mirror of a physical entity—offers a compelling conceptual and operational model for reimagining evaluation as *algorithmic reflexivity*.

The Digital Twin as a Model of Co-Evolution

Originally rooted in engineering, the Digital Twin serves as a high-precision, continually updated simulation of a physical system. In healthcare, it depicts a patient through an algorithmically driven identity that constantly learns from diverse data streams such as genomics, sensor outputs, and medical records.[10].This ongoing synchronization between biological data and computational models enables predictive analytics, adaptive treatments, and dynamic optimization of health outcomes.

Philosophically, the Digital Twin extends beyond its technical origins. It reflects a relational ontology—viewing identity, agency, and governance as co-created by humans, algorithms, and institutions.[11]. Consequently, it shifts healthcare from a fragmented, physician-centered approach to a cybernetic ecosystem of care, where feedback loops constantly influence diagnosis, treatment, and trust.

This human–machine co-creation model offers a compelling analogy for future research evaluation. Just as the Digital Twin forecasts patient health in real time, a Reflexive Evaluation Framework can simulate and refine the health of a research ecosystem—monitoring its methodological robustness, ethical standards, and social impact in a dynamic balance.

Toward the Reflexive Evaluation Framework (REF)

Building on the Digital Twin concept, the REF redefines evaluation as a dynamic digital ecosystem. It functions not as an external audit tool but as an internal intelligence that develops alongside the research it assesses.[12]. The REF reflects the three-dimensional structure of the Digital Twin, consisting of:

1. Identity Construction — Establishing and continually updating the research system's "ontology" (its goals, stakeholders, and context).

2. Data Agency and Learning — Incorporating real-time data from various sources to enable ongoing feedback and methodological adjustments.

3. Adaptive Governance and Ethics — Implementing flexible oversight mechanisms that adapt to new ethical, social, and scientific issues.

These three components interact in a reflexive cycle of inquiry and adaptation. The REF "learns" from the research system it monitors, much as a Digital Twin learns from the patient

it models. This method keeps evaluation both rigorous and responsive, maintaining a balance between accountability and flexibility—a vital duality often missing in traditional frameworks.

The Reflexive Logic of the Digital Twin

From an Empirics-First (EF) perspective, the Digital Twin (DT) is more than a metaphor; it serves as a shifting methodological lens that facilitates the assessment of real-world data. This does not imply that the REF necessitates prior application of theoretical principles before evaluation. Instead, within the DT's methodological framework, the Reflexive Evaluation Framework (REF) advances through an inductive observation process. Four levels of reflexivity—Continuous Synchronization, Simulation and Prediction, Transparency and Explainability, and Co-Evolutionary Adaptation—guide this process by transforming evaluation from a static analysis into a continuous, dynamic activity evolution.

Table 2. Conceptual Parallels Between Digital Twin Principles and the Reflexive Evaluation Framework (REF)

Digital Twin Principle	**Analogous Principle in Reflexive Evaluation Framework (REF)**
Continuous Synchronization — Real-time data integration enables continuous model updates. [1]	Continuous feedback and learning loops update research evaluations dynamically as evidence evolves. [13]
Simulation and Prediction — The DT anticipates system responses before interventions. [14]	The REF simulates potential impacts of research and policy decisions before implementation. [15]
Transparency and Explainability — The DT's algorithms must be auditable and interpretable to sustain trust.[16]	The REF embeds algorithmic transparency, stakeholder participation, and interpretability as ethical imperatives. [17]
Co-Evolutionary Adaptation — The DT evolves in tandem with patient and environmental conditions. [18]	The REF co-evolves with scientific paradigms, data ecosystems, and societal needs. [19]

By applying these principles, the REF shifts from just an evaluative endpoint to a knowledge-producing system that can reflect on, model, and steer the processes it examines. As shown in Table 2, this transforms evaluation from a static analysis into a dynamic ecosystem of reflexive learning.

Integrating Evaluation, Ethics, and Governance

Traditional evaluation frameworks often isolate methodology from morality—treating ethical oversight as external regulation. The REF collapses this divide by embedding adaptive ethics into its architecture. Borrowing from the Digital Twin's *adaptive governance* principles, it establishes:

- **Dynamic Consent and Stakeholder Inclusion:** Continuous negotiation of participation and transparency in evaluation processes.
- **Algorithmic Accountability:** Systematic auditing of evaluative metrics, biases, and interpretive assumptions.
- **Equity-by-Design:** Ensuring that inclusivity and representational diversity are built into data collection and evaluative models from inception.
- **Learning Ethics:** Treating ethical oversight as iterative and reflexive—responsive to new data, feedback, and context.

In this sense, the REF mirrors the *ethical metabolism* of the Digital Twin: a self-regulating system that sustains legitimacy by continuously aligning its internal operations with evolving moral and epistemic norms. Consistent with the Empirics-First (EF) paradigm, this metabolism emerges inductively from real-world observation rather than theoretical prescription.

The REF as a Socio-Technical Ecosystem

The Digital Twin and REF both challenge the traditional divide between "subject" and "system." In the REF, the researcher, evaluator, and institution are all part of a shared feedback ecosystem. This makes the framework socio-technical—combining algorithmic infrastructure with an ethical community.[20] By incorporating data analytics, stakeholder participation, and participatory governance, the REF functions simultaneously as:

1. A Cognitive Model — illustrating how research knowledge develops and validates.[21]

2. An Ethical System — integrating reflexive accountability into decisions.[22]

3. An Institutional Mechanism — aligning policies, incentives, and feedback loops across different organizational levels.[23]

These interconnected layers turn evaluation into an ecosystem of co-production, where human judgment and algorithmic reasoning work symbiotically.

Reflexivity as the Epistemic Core

The core concept of the REF is reflexivity—the ability of a system to reflect on its own learning process. This sets adaptive evaluation apart from traditional oversight methods. It means that knowledge is not just created but also continuously analyzed, leading to deeper insights into how knowledge is organized, limited, and shaped by context.[24] In this perspective, evaluation acts as a form of meta-intelligence: a persistent dialogue between objectivity and subjectivity, technical accuracy and ethical considerations. Consequently, the REF functions as both an analytical tool and a moral guide—ensuring that the development of knowledge remains open, inclusive, and responsible to its wider human and institutional contexts.

Conceptual Diagram: The Reflexive Digital Twin Model

Figure 1 below illustrates the relationship between the Digital Twin concept and the Reflexive Evaluation Framework (REF).

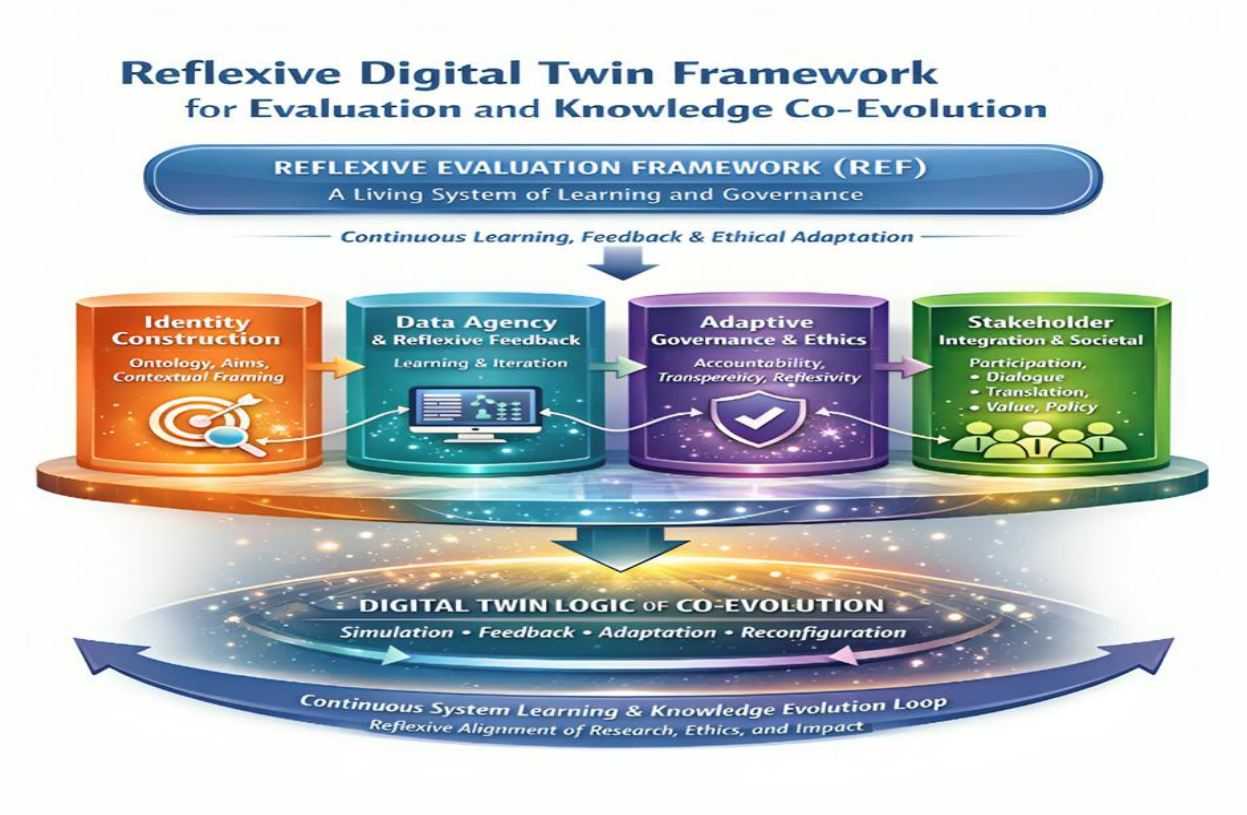

Fig 1: Reflexive Digital Twin Framework for Evaluation and Knowledge Co-Evolution

Diagram Explanation

This diagram depicts the REF as a Digital Twin of Research Systems—a continuously evolving structure that reflects and enhances its own epistemic and ethical processes.

1. Base Layer – Digital Twin Co-Evolution Logic

The lowest layer shows the core logic of the Digital Twin—simulation, feedback, adaptation, and reconfiguration—mirroring the cycle of ongoing improvement seen in intelligent systems and human–machine learning architectures.

2. Middle Layer – Reflexive Evaluation Core

Within the REF, four interconnected subsystems operate:

- Identity Construction: defines research goals, ontology, and context.
- Data Agency & Reflexive Feedback: facilitates continuous learning from data and outcomes.
- Adaptive Governance & Ethics ensures accountability, fairness, and transparency.
- Stakeholder & Societal Impact: translates insights into policy, practice, and value creation.

3. Top Layer – Reflexive Integration and Learning Loop

This layer shows how continuous feedback flows upward, ensuring evaluation aligns with ethical governance and societal relevance, transforming research into a dynamic, co-evolving system.

This diagram depicts the REF as a Digital Twin of research—a system that continuously monitors, models, and improves its epistemic and ethical functions through co-evolutionary feedback. It integrates technical accuracy with moral reflexivity, forming a balanced structure of learning and legitimacy.

From Evaluation to Evolution

Integrating the Digital Twin concept with knowledge system assessment, the Reflexive Evaluation Framework (REF) marks a significant epistemic shift. It shifts the view of evaluation from merely judgment to a co-learning process—a form of self-modelling that helps systems stay relevant, resilient, and responsible. By incorporating reflexive feedback, ethical flexibility, and stakeholder engagement, the REF transforms evaluation into a dynamic epistemology.[25]. This approach enables effective handling of uncertainty while maintaining transparency, inclusivity, and shared accountability. Essentially, it applies the logic of intelligent systems to ethical inquiry: an evaluation that adapts as the world it aims to understand changes.

Methodological Implications: Operationalizing the Reflexive Evaluation Framework (REF)

Methodology: Open-Ended Scoping Review and Conceptual Synthesis

This research employs an open-ended scoping review combined with conceptual synthesis to implement the Reflexive Evaluation Framework (REF) in the realm of Digital Twin (DT) research and marketing science. The goal is not to validate a specific theory but to explore how AI-driven simulation, adaptive governance, and data agency transform healthcare ecosystems and marketing research paradigms.

The scoping review is particularly suitable due to the innovative and interdisciplinary nature of the Digital Twin concept. Instead of limiting the inquiry with hypothesis-driven inclusion criteria, the review aims to encompass the full scope of knowledge across various fields—from biomedical engineering and ethics to health informatics, marketing analytics, digital transformation, and stakeholder co-creation.[26]

Following PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses–Scoping Review) standards, the process emphasizes transparency, reproducibility, and reflexivity, moving away from a theory-first approach. Instead, the literature serves as a discovery space—a reflexive environment where empirical results and conceptual ideas develop through iterative synthesis.

This methodology considers the Digital Twin not just as a technological tool but as a socially embedded metaphor for learning systems, aligning with the REF's epistemic aim of viewing research as a dynamic, adaptive living system.

Research Process: Breadth, Depth, and Reflexivity

The empirical scoping and synthesis process was built around three meta-principles, all aligned with the Reflexive Evaluation Framework (REF). During the scoping review, the REF offered a structured approach to organizing and interpreting the diverse literature emerging at the intersection of Digital Twin (DT) systems, marketing analytics, and ethical AI governance.[27]. The initial DT–REF mapping created conceptual analogies that guided the coding and data extraction, emphasizing co-evolution, transparency, and continuous learning. Building from this, the analysis progressed iteratively as new patterns emerged within the corpus. Guided by the principles of breadth, depth, and reflexivity, the review not only described the data but also uncovered underlying constructs and epistemic tensions across different fields—reflecting an Empirics-First (EF) approach in which theoretical insights emerge inductively from observed empirical regularities rather than from predetermined frameworks. Table 3 demonstrates how these principles were put into action to interpret interdisciplinary evidence and integrate insights in the Digital Twin–Marketing area, connecting empirical mapping with conceptual theory development.

Table 3. Core Analytical Principles of the Reflexive Evaluation Framework (REF) and Their Application in the Digital Twin–Marketing Nexus

Principle	Definition and Role in the REF	Example in Digital Twin–Marketing Nexus
Breadth	Capture a diverse, multi-disciplinary range of sources to identify emergent constructs beyond disciplinary silos.	Mapping DT literature across healthcare AI, marketing analytics, and ethics to understand shared challenges of data agency and stakeholder trust.
Depth	Interrogate the *latent assumptions* and *ontological framings* underlying how systems, actors, and identities are represented.	Analyzing how “patients,” “consumers,” or “decision-makers” are algorithmically defined and governed within twin-driven ecosystems.

Reflexivity	Continuously re-evaluate one's analytical stance as the literature and field evolve, ensuring responsiveness to emerging patterns.	Revising coding frames mid-review as novel constructs (e.g., "algorithmic empathy," "adaptive governance") surfaced across data streams.

Unlike traditional literature reviews that set boundaries beforehand, this approach was iterative and exploratory. Each cycle of analysis influenced the next, creating feedback loops like the Digital Twin's co-evolutionary process — where understanding of the system improves through simulation, feedback, and adjustment. Subsequent rounds of literature searching, coding, and conceptual clustering were conducted as new insights reshaped key categories (e.g., data agency, stakeholder co-production, adaptive ethics). In this way, the methodology became reflexive — serving both as an analytical tool and as a focus for meta-evaluation.

Search Strategy and Data Sources (Journal of Marketing–Aligned Approach)

To ensure breadth, rigor, and alignment with marketing research standards, the search strategy followed multi-database triangulation, integrating both *disciplinary relevance* and *cross-sectoral convergence*:

- **Databases Searched:** Web of Science, Scopus, ABI/Inform, PubMed, and Google Scholar.
- **Keywords and Boolean Combinations:**

  "Digital Twin" AND ("healthcare" OR "patient" OR "AI-driven simulation")

  "Digital Twin" AND ("marketing analytics" OR "customer modelling" OR "digital representation")

  "adaptive governance" OR "algorithmic accountability" OR "data agency"

  "reflexive evaluation" OR "stakeholder learning" OR "ethical AI systems"

- **Temporal Scope:** 2010–2025, capturing both conceptual origins and recent AI-driven expansions.
- **Inclusion Logic:** Conceptual, empirical, or review papers contributing to discussions of DT applications, ethics, or co-evolutionary learning in socio-technical systems.

Screening was conducted in two iterative waves:

First, exploratory mapping to identify key research clusters, including clinical DT models, AI governance, and marketing analytics.

Second, analytical refinement of code to capture emergent constructs relevant to the REF, including feedback integrity, adaptive trust systems, and multi-stakeholder co-learning. This two-phase approach balanced openness to new concepts with rigorous, systematic interpretation.

Methodological Integration: From Data to Reflexive Modelling

Operationalizing the Reflexive Evaluation Framework involved translating scoping insights into a conceptual architecture guided by three methodological strategies:

1. Iterative Conceptual Coding:[28]

Using qualitative synthesis tools such as NVivo, literature segments were coded for constructs related to REF dimensions—such as identity construction, data agency, adaptive governance, and stakeholder integration. These codes were refined through iterative consensus sessions, embodying the REF's feedback–adaptation–reconfiguration loop.

2. Cross-Domain Convergence Mapping:[29]

Thematic matrices were created to illustrate how similar mechanisms appeared across different domains.

For example, "predictive modelling" in healthcare corresponded conceptually to "forecasting

consumer behavior" in marketing—both addressing issues of data ethics, transparency, and co-production.

3. Conceptual Simulation (Reflexive Modelling):[30]

Inspired by Digital Twin simulation logic, emergent constructs were modelled in "what-if" scenarios (e.g., What occurs when stakeholder feedback is incorporated as algorithmic input?).

This simulative reasoning strengthened the theoretical foundation of the REF, ensuring that its methodology reflected co-evolutionary thinking.

Reflexive Evaluation as Methodological Ethos

The REF extends beyond mere technique to embody an epistemic ethos in which evaluation and research function as ongoing, learning-driven processes.

Key implications for marketing and healthcare research include:

• Feedback as Data: Evaluation metrics such as engagement rates and adoption indices should be viewed as real-time learning signals rather than fixed performance scores. [31]

• Dynamic Consent Models: Inspired by healthcare ethics, marketing research can implement adaptive consent frameworks that allow participants to renegotiate data usage as studies develop.[32]

• Participatory Evaluation Protocols: Stakeholders—patients, consumers, and partners—[33]become co-analysts instead of just data sources, reflecting Digital Twin participatory logic. [34]

• Ethical Co-Learning: Evaluation should explicitly model how insights influence both systems and individuals, creating "value loops" across research and practice. [33]

Overall, the REF provides a meta-methodological perspective: a system for designing, executing, and refining research in real time, rooted in transparency, flexibility, and ethical awareness.

Implications for Marketing Research Practice

Applying this **reflexive and twin-inspired methodology** transforms marketing research in three major ways:

1. Transitioning from Rigid Design to Adaptive Inquiry, research evolves into a dynamic co-learning process that continually updates hypotheses based on real-world data and stakeholder feedback.
2. Moving from Metrics to Meaning, evaluation shifts from purely performance indicators like ROI and customer satisfaction to encompass ethical integrity, trust development, and societal impact.
3. Evolving from Audience to Co-Producer, stakeholders are reconceptualized as active epistemic agents who co-create meaning through participation and interaction with algorithms.

This connects marketing to contemporary movements such as stakeholder capitalism, responsible AI, and adaptive governance, thereby maintaining its relevance and credibility in the digital age. The methodological framework presented here adopts the Reflexive Evaluation Framework (REF) as both a research approach and an evaluative philosophy. By combining comprehensive scoping reviews, in-depth conceptual synthesis, and reflexive learning, it enables marketing and healthcare researchers to emulate the co-evolutionary intelligence of the Digital Twin. In this process, methodology itself becomes an active agent in the development of knowledge—constantly adapting to complexity, improving transparency, and promoting a shared culture of ethical innovation across research, industry, and society.

Results: Mapping the Knowledge Landscape of the Digital Twin and Reflexive Evaluation Framework

Overview

Guided by an Empirics-First (EF) approach, the scoping review took an open-ended, data-driven method to explore how reflexivity appears in the literature on Digital Twin marketing and governance. Instead of starting with preset hypotheses, it aimed to discover empirical patterns and conceptual conflicts within the body of research. Following PRISMA-ScR standards, literature searches spanned interdisciplinary databases—*Web of Science, Scopus, PubMed,* and *Google Scholar*—covering the period 2010–2025. This process emphasized transparency, reproducibility, and depth, ensuring that the subsequent conceptual synthesis rested on robust empirical foundations. Table 4 provides an overview of the review process, including the search criteria, record counts, and the final dataset used to develop the Reflexive Digital Twin Marketing (RDTM) theory.

Table 4: Empirics-First Scoping Review Process and Record Flow Across PRISMA-ScR Stages

Stage	Process Description	Number of Records	Notes
Identification	The database search was performed across Web of Science, Scopus, PubMed, and Google Scholar using Boolean combinations of key terms such as “Digital Twin,” “healthcare,” “marketing analytics,” “adaptive governance,” “data agency,” and “reflexive evaluation.”	**742**	Search range: 2010–2025; peer-reviewed studies published in English.

Screening	Duplicates have been removed; abstracts were screened to ensure relevance to Digital Twin applications, marketing systems, and socio-technical perspectives.	**528**	214 duplicates and non-English sources were excluded.
Eligibility	Full-text articles were evaluated for their conceptual depth, empirical rigor, and relevance to the Reflexive Evaluation Framework (REF) dimensions.	**145**	90 excluded: engineering-only focus (n = 37), insufficient ethical or theoretical content (n = 28), or inaccessibility or non-peer-reviewed (n = 25).
Inclusion	Studies meeting the inclusion criteria were conceptually synthesized across four REF dimensions: Continuous Synchronization, Simulation and Prediction, Transparency and Explainability, and Co-Evolutionary Adaptation.	**55**	Final corpus distributed across domains: Marketing (n = 20), Healthcare (n = 18), and Governance and Ethics (n = 17).

The systematic scoping review revealed not only the breadth of emerging research on Digital Twin applications but also its fragmentation across disciplines. By filtering studies through the Reflexive Evaluation Framework (REF), conceptual patterns emerged around four core dimensions—*identity construction, data agency, adaptive governance,* and *stakeholder integration*. These dimensions formed the empirical structure from which the Reflexive Digital Twin Marketing theory evolved, exemplifying the EF principle that data precede abstraction, and that theory develops from patterns observed in the real world rather than from preexisting models.

PRISMA-ScR Flow Diagram

In keeping with the Empirics-First (EF) perspective, this stage of the study marks the point at which empirical discovery began to converge toward conceptual coherence. Rather than treating the scoping review as a procedural step, it was framed as a learning system—a reflexive

process through which patterns, anomalies, and thematic intersections in the Digital Twin (DT) literature began to reveal their theoretical significance. The review's iterative filtering helped surface a core set of studies situated at the nexus of marketing analytics, data ethics, and adaptive governance, marking the empirical ground from which Reflexive Digital Twin Marketing (RDTM) theory could evolve. Figure 2 visualizes this progression, illustrating how systematic evidence selection became a generative stage of theoretical abstraction rather than a purely methodological exercise.

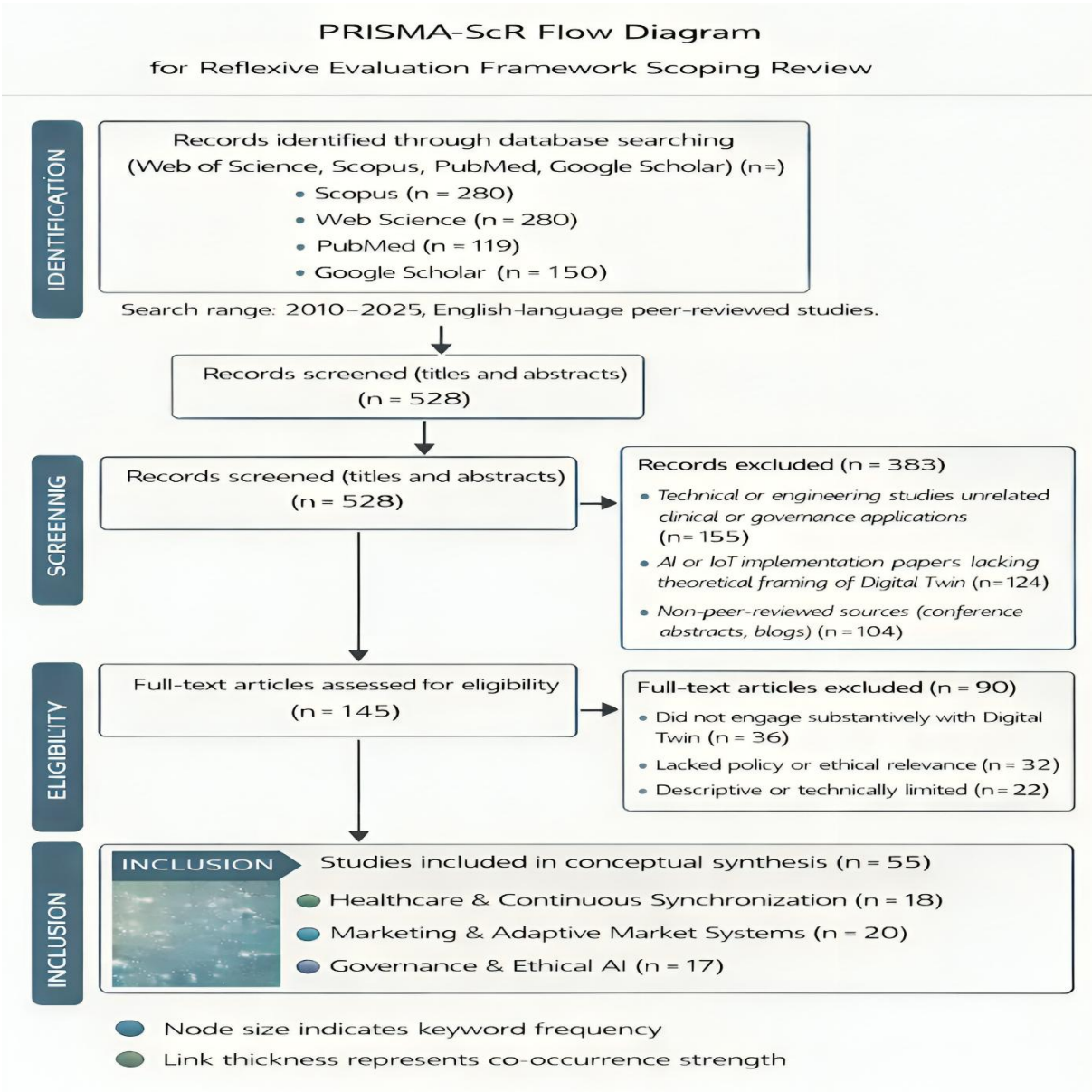

Fig 2: PRISMA-ScR diagram representing the Literature Search

The body of literature, shown in Figure 2, derives from 55 studies identified through a transparent, multi-stage scoping review. These studies—covering health innovation (n = 18), marketing and data-driven management (n = 20), and governance and ethical AI frameworks (n = 17)—represent the point at which disciplinary conversations began to converge around shared concerns such as data agency, transparency, and adaptive learning. Instead of defining a fixed theoretical boundary, this synthesis revealed an evolving network of reflexive concepts linking the technical, managerial, and ethical aspects of Digital Twin research. In line with the Empirics-First (EF) approach, theory was allowed to emerge from interaction and pattern recognition rather than being imposed by prior models. The studies served as conceptual anchors, guiding subsequent inquiry through VOS co-occurrence mapping and the Reflexive Evaluation Framework (REF) synthesis, which uncovered the structural interdependencies that underpin the Reflexive Digital Twin Marketing (RDTM) theory.

From Systematic Mapping to Theoretical Integration

The PRISMA-ScR flow diagram (Figure 2) illustrates the transparent and systematic process used to identify 742 records and refine them through sequential screening and eligibility stages to a final set of 55 studies included in the conceptual synthesis. This corpus, spanning healthcare innovation, marketing analytics, and governance ethics, served as the empirical foundation for the Empirics-First (EF) synthesis that guided the study's theoretical development. Building on this dataset, the VOS co-occurrence mapping (Figure 3) visualizes the intellectual structure of the literature, revealing four primary thematic clusters: (1) Healthcare and Continuous Synchronization, (2) Marketing Analytics and Adaptive Intelligence, (3) Reflexive Governance and Ethical AI, and (4) Societal Co-Evolution and Stakeholder Ecosystems. Together, these clusters depict how diverse disciplinary streams converged on the concept of reflexivity—a dynamic principle that links data, decision-making, and adaptive learning across socio-technical contexts.

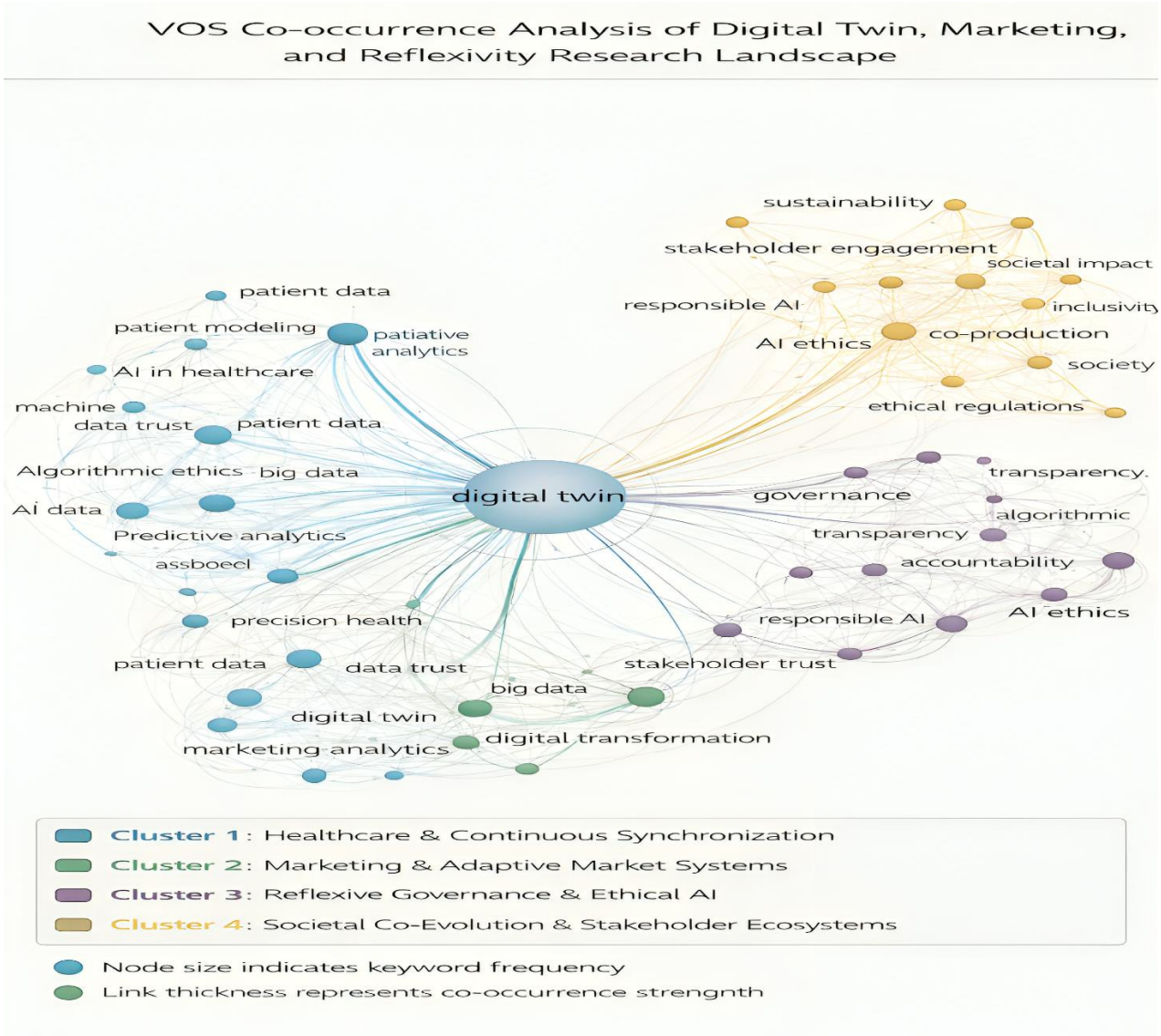

Fig 3: Mapping of the Reflexive Evaluation Framework (REF) Scoping Review using integrated PRISMA-ScR and VOS methods.

Figure 3 illustrates the VOS co-occurrence mapping, showing the intellectual structure of the 55 studies in the conceptual synthesis. The analysis identifies four interconnected thematic clusters that form the core principles of Reflexive Digital Twin Marketing (RDTM) theory.

- Cluster 1 (Healthcare and Continuous Synchronization) emphasizes the ongoing synchronization of biological, behavioral, and environmental data, demonstrating how real-time adaptation and patient involvement enhance reflexive intelligence systems.
- Cluster 2 (Marketing Analytics and Adaptive Intelligence) embodies the application of feedback logics within market systems, where data-driven learning, predictive modelling, and personalization facilitate the co-evolution of firms and consumers.
- Cluster 3 (Reflexive Governance and Ethical AI) emphasizes the importance of algorithmic transparency, accountability, and explainability in maintaining stakeholder trust and ensuring ethical legitimacy within adaptive ecosystems.
- Cluster 4 (Societal Co-Evolution and Stakeholder Ecosystems) embodies the macro-level interaction among technology, institutions, and human values, with reflexivity appearing as shared learning across markets and societies.

These clusters collectively show how various disciplinary insights converge around reflexivity as a systemic feature—evolving Digital Twin research from a mere technical approach to a socio-ethical and managerial framework vital to adaptive marketing theory.

Thematic Clustering and Conceptual Convergence

Following the Empirics-First (EF) approach advocated by P. N. Golder et al. (2023), this stage of the scoping review and VOS co-occurrence moves from descriptive mapping to empirical synthesis. Instead of beginning with a fixed theoretical model, patterns within the literature were allowed to surface inductively, revealing recurring dimensions of reflexivity across digital twin, marketing, and governance studies. Using the Reflexive Evaluation Framework (REF) as an interpretive scaffold, four empirically grounded dimensions—*Identity Construction*, *Data Agency and Reflexive Feedback*, *Adaptive Governance and Ethics*, and *Stakeholder Integration and Societal Impact*—were identified. Table 5 summarizes how these dimensions manifested

across the corpus, illustrating how data-rich marketing systems increasingly exhibit reflexive characteristics: learning, adaptation, and ethical self-regulation. This synthesis thus represents the pivot point between empirical observation and conceptual abstraction within the Reflexive Digital Twin Marketing (RDTM) framework.

Table 5. Reflexive Evaluation Framework (REF) Dimensions Identified Through the Scoping Review and Their Corresponding Insights Across the Digital Twin–Marketing Literature.

REF Dimension	Illustrative Focus in Literature	Key Insights Identified
Identity Construction	Framing of DT as a representational entity in healthcare and marketing; definitions of "patient," "consumer," and "data subject."	DTs redefine subjectivity and accountability; move from passive representation to active co-production.
Data Agency & Reflexive Feedback	Machine learning, predictive analytics, and feedback loops; co-creation of meaning between humans and algorithms.	Continuous data calibration transforms both system and user identity.
Adaptive Governance & Ethics	Transparency, algorithmic accountability, privacy, and adaptive regulatory models.	Emergence of ethics-by-design and self-regulating data infrastructures.
Stakeholder Integration & Societal Impact	Cross-sectoral applications in marketing, public health, and digital ecosystems.	Multi-stakeholder collaboration fosters legitimacy and learning in socio-technical systems.

Through the Empirics-First lens, these dimensions reveal a shift from descriptive observation to conceptual emergence. Rather than validating predefined constructs, the analysis surfaced *new relational patterns*—for example, how algorithmic personalization redefines consumer identity, or how ethics-by-design becomes an adaptive governance function. These empirical regularities collectively form the scaffolding for Reflexive Digital Twin Marketing (RDTM)

Theory, demonstrating how reflexive feedback loops link technological design, market learning, and stakeholder legitimacy. In the EF tradition, theory here does not precede the data; it evolves *from* data, representing an iterative co-creation between evidence and conceptual understanding.

Conceptual Patterns and Cross-Domain Insights

1. Healthcare–Marketing Convergence: Both fields highlight the importance of real-time personalization and predictive modelling as shared objectives. Healthcare digital twins (DTs) mimic physiological processes, while marketing DTs imitate consumer behaviour, illustrating similar ethical challenges related to data autonomy and algorithmic transparency.

2. Emergent Reflexivity: Across various studies, reflexivity appears both as a methodological approach—where researchers revisit assumptions during data collection—and as an ethical perspective—recognizing the social implications of digital modelling. This finding underscores the REF's role as a tool for meta-learning and adaptive governance.

3. Gaps and Underexplored Frontiers: Despite notable technological progress, the literature reveals gaps in comprehensive frameworks that link technical simulation, ethical oversight, and stakeholder learning—precisely the area the REF aims to address conceptually.

Synthesis Summary

The scoping review results collectively demonstrate that, while the Digital Twin has been extensively studied as a technological and engineering artifact, its social, ethical, and reflexive dimensions remain fragmented across disciplines.

By synthesizing findings through the Reflexive Evaluation Framework (REF), this study shows how evaluation itself can evolve into a digital-ethical twin of inquiry—learning from data, adapting through feedback, and co-producing meaning with stakeholders.

The PRISMA-ScR diagram not only documents the transparency of the research process but also metaphorically mirrors the twins' learning loop—identifying, filtering, and refining knowledge through reflexive adaptation.

Visual Summary: Integrative PRISMA–REF Alignment

To visually illustrate this dual process, the PRISMA flow and REF learning cycle can be seen as two interconnected systems:

• PRISMA Flow → guarantees transparency and reproducibility in knowledge selection.

• REF Loop → promotes reflexivity and flexibility in interpreting knowledge.

Together, they create a cybernetic evaluation framework that learns, filters, and continually evolves.

Fig 4. Reflexive Digital Twin Marketing (RDTM) Theory Framework

Figure 4 presents the Reflexive Digital Twin Marketing (RDTM) Theory, viewing markets as co-evolving ecosystems of digital and human elements. The framework combines Digital Twin principles—such as simulation, feedback, adaptation, and reconfiguration—with marketing system dynamics, including data agency, stakeholder interactions, ethical governance, and societal impacts.

At its core is the Reflexive Evaluation Loop, which signifies ongoing learning among marketing systems, digital representations, and stakeholder ecosystems.

Four interlinked domains form the theoretical architecture:

1. Identity Construction – outlines the roles of digital and market participants, ontologies, and contextual goals.
2. Data Agency & Reflexive Feedback – describes adaptive learning processes that allow systems to self-adjust using both algorithmic updates and human insights.
3. Adaptive Governance & Ethics – incorporates accountability, transparency, and moral considerations into marketing decision-making processes.
4. Stakeholder & Societal Co-Production – guarantees that marketing intelligence and innovations support social welfare, sustainability, and inclusivity.

The outer loop represents market reflexivity, where input from practice—such as consumer feedback, ethical evaluations, and societal discussions—adjusts strategy, governance, and theory. This co-evolutionary process makes marketing a dynamic, learning system that continuously adapts and evolves through reflection.

Discussion:

Real-World Relevance: Marketing in the Age of Intelligent Representation

The findings highlight a significant transformation in how marketing systems impact human welfare, stakeholder involvement, and social responsibility. Originally limited to engineering and healthcare, the Digital Twin (DT)[35], now serves as a model for capturing dynamic market realities—continuously mirroring consumers, brands, and institutions through data-driven interactions. In real-world marketing environments—from predictive analytics to experiential retail—the DT approach enables simulated marketing scenarios, allowing strategy testing via digital representations before implementation. This shifts marketing from mere descriptive analysis to proactive co-creation, directly supporting human-centered welfare, ethical personalization, and sustainability. Despite ongoing debates—ranging from utopian efficiency to dystopian surveillance—integrating DT with Reflexive Evaluation Frameworks (REFs) clarifies that DT systems can enhance societal accountability in marketing when embedded within transparent, reflexive, and participatory evaluation. Consequently, marketing shifts from merely promoting consumption to fostering responsible co-evolution among firms, markets, and society.

Literature Evolution: From Predictive Analytics to Reflexive Intelligence

The review shows that while current marketing literature is data-rich in analytics and customer modelling, it lacks reflexivity and is divided into silos. Existing frameworks in AI marketing, CRM, and digital transformation mainly focus on prediction and personalization, but they rarely explore how these systems learn from their own processes.[36]. By integrating DT's feedback-driven structure with marketing theory, this research broadens the field's scope in two significant ways:

1. **Moving from Linear Causality to Reflexive Systems:**

   Traditional marketing views causality as unidirectional—the firm influencing the consumer.[37]. The RDTM perspective introduces recursive causality, in which both consumers and firms continuously adapt in real time through data feedback, creating learning markets rather than static target markets.

2. **Moving from Descriptive Models to Epistemic Simulations:**

   Marketing models transition from static predictors, such as regression or segmentation,[38] to dynamic epistemic twins—systems that simulate behavioural responses, ethical considerations, and societal impacts before strategy deployment.

   This shift aligns marketing more closely with fields such as cybernetics, systems theory, and digital ethics, which see markets as living, reflexive systems rather than mere mechanical exchanges.

Research Process: Reflexivity as a Methodological Ethic

The study's scoping review and conceptual synthesis employed an open-ended, agenda-free approach, enabling theories to develop naturally through interdisciplinary collaboration. Led by the REF, the process balanced scope, depth, and reflexivity, uncovering the dynamic interaction between technology, ethics, and market systems.

- Breadth: Including 742 records across healthcare, informatics, and marketing ensured exposure to diverse epistemic traditions.
- Depth: By interrogating latent meanings—e.g., who the "digital consumer" is and how "value" is algorithmically constructed—the review uncovered deep parallels between patient and consumer modelling.

- Reflexivity: The iterative coding and theory-building process mimicked the Digital Twin's simulation loop, continuously adjusting theoretical boundaries in response to emerging insights.

This methodological reflexivity resulted in a research design that reflects on itself, embodying the same logic as the phenomenon being examined.

Research Outcomes: Discovering a Reflexive Regularity in Marketing Systems

The conceptual synthesis reveals an empirical pattern across marketing and healthcare digital twin (DT) applications: systems that incorporate reflexive feedback loops tend to achieve higher adaptive performance and gain greater stakeholder trust. In marketing, this concept is embodied by predictive ethics—the capacity of systems not only to forecast consumer preferences but also to foresee ethical and societal impacts. The RDTM theory, therefore, emphasizes reflexivity as a key factor influencing performance, connecting economic and social outcomes. Practical implications for marketing include:

• Integrating digital twin simulations into strategic decision-making to predict consumer attitudes and societal reactions.

• Creating real-time dashboards that combine marketing KPIs with metrics for ethics, sustainability, and inclusion.

• Using dynamic consent features that enable customers to co-manage their digital representations. These developments broaden marketing's role from solely predicting behaviour to fostering participatory sensemaking, where value is co-created, validated, and ethically guided.

Robustness and Generalizability: Learning from Failure and Cross-Context Convergence

The PRISMA-ScR synthesis shows that while DT applications vary across sectors, their success consistently depends on reflexive governance mechanisms. Failed implementations (e.g., overfitted predictive models or non-transparent algorithms) provide essential learning opportunities—illustrating that robustness in marketing models arises not from control but from adaptability.

The RDTM framework demonstrates generalizability across domains:

• In healthcare, DTs optimize clinical outcomes through adaptive modelling.[39]

• In marketing, they optimize consumer engagement through ethical learning.[40]

• In governance, they sustain institutional legitimacy through transparency and participatory oversight.[41]

Thus, reflexivity itself emerges as a meta-principle of generalizability—a foundation for transdisciplinary knowledge transfer and theory development.

Presentation and Transparency: Explaining the Discovery Process

This study's significant contribution is its clear presentation of the Empirics-First (EF) discovery process, which shifts theory development from a fixed goal to an ongoing, reflective learning system. As shown in Figure 5, the process involves iterative cycles that combine empirical data, conceptual insights, and practical use. Each cycle concludes by revisiting the study's initial aims: improving the practical relevance of marketing research, tackling ethical and epistemic issues in algorithmic decision-making, and incorporating a systemic mechanism—the Reflexive Digital Twin (RDT)—that continuously aligns research with societal wellbeing. This repeated process exemplifies the Reflexive Evaluation Framework (REF) principle that understanding arises through active engagement, not just following set rules. By illustrating these feedback loops visually and narratively, the study showcases

methodological integrity, strengthening both the credibility of its approach and the reliability of its findings.

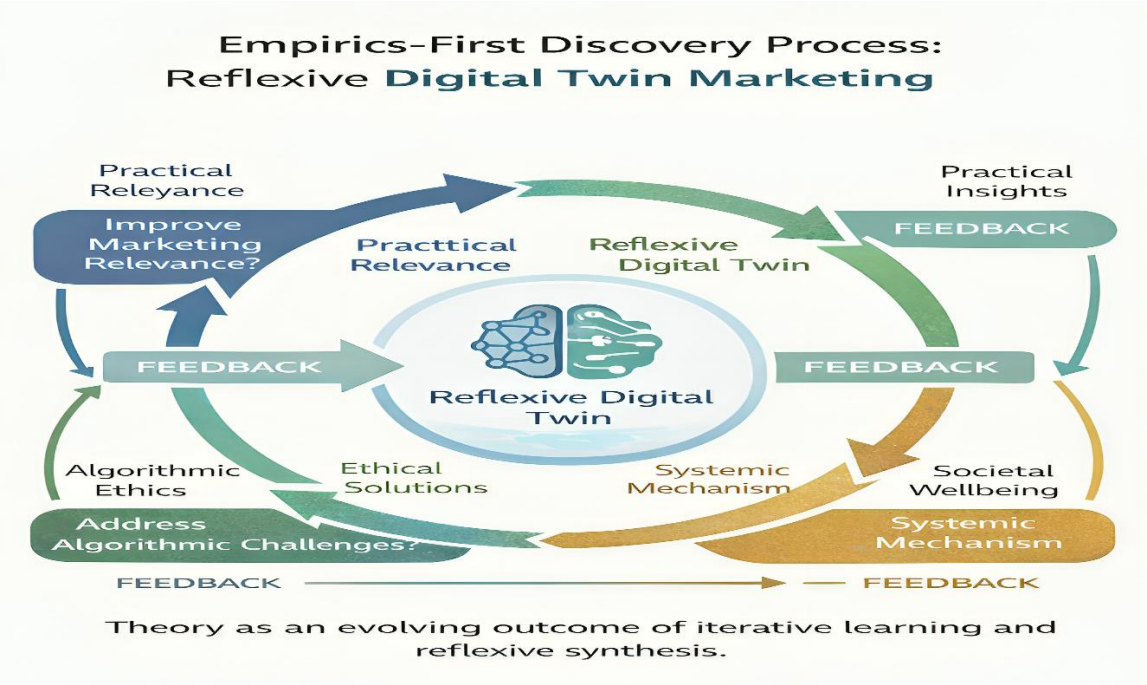

Figure 5. Reflexive Learning Loops in Developing Reflexive Digital Twin Marketing (RDTM) Theory

Emergence of Reflexive Digital Twin Marketing (RDTM) Theory

This integration introduces a new marketing theory, Reflexive Digital Twin Marketing (RDTM). It is described as:

A systemic framework conceptualizes markets as co-evolving digital–human ecosystems, with reflexive feedback loops that continuously align firm strategies, stakeholder values, and societal outcomes.

Core Propositions:

1. Markets as Living Systems — Marketing isn't just linear cause-and-effect but involves adaptive systems that can learn and self-correct.

2. Reflexivity as Market Intelligence — The key marketing insight is less about prediction accuracy and more about adaptive learning ability.

3. Ethics as System Architecture — Ethical considerations and participatory evaluation are integral, structural parts of marketing performance.

4. Co-Simulation as Value Creation — Companies and stakeholders collaboratively create value by iteratively modelling and learning from each other's digital twin representations.

New Concepts Introduced:

• Digital Reflexivity: The ability of marketing systems to recognize and correct their own biases.

• Ethical Feedback Elasticity: How much can market strategies change in response to moral, social, and stakeholder feedback?

• Value Coherence Index: A combined metric measuring the alignment of marketing outcomes, ethics, and societal impacts.

Collectively, these ideas advance marketing theory by framing it as a cybernetic system that merges human-centred strategies with machine learning.

Closing the Loop: From Reflexivity to Responsibility

This discussion situates the Reflexive Digital Twin Marketing (RDTM) Theory within the broader shift of marketing toward a responsible knowledge system. The Digital Twin evolves from a modelling tool into a moral and epistemic partner guiding sustainable marketing futures. By connecting real-world relevance, integrating the literature, employing adaptive research methods, and synthesizing reflexive theory, this study completes the methodological cycle envisioned by the Reflexive Evaluation Framework (REF): turning evaluation into evolution and research into a dynamic system of societal learning.

Conclusion and Implications

Integrative Summary

This research advances an Empirics-First (EF) approach to developing theories in marketing, resulting in the Reflexive Digital Twin Marketing (RDTM) Theory. Using a PRISMA-ScR-guided scoping review and VOS co-occurrence mapping of 742 studies from healthcare, marketing analytics, and ethical governance, the study highlights the common logic of simulation, feedback, and co-evolution that characterizes current data-driven marketing systems. The findings indicate that marketing, similar to intelligent healthcare or adaptive industrial systems, is shifting from predictive management to reflexive learning—where feedback from markets, stakeholders, and technologies continuously influences strategy and understanding.

By viewing marketing as a dynamic, data-enabled ecosystem, the RDTM framework redefines the discipline's epistemic base: marketing is not just about influencing behavior but also about learning from the behaviors it shapes.[42]

The RDTM Theory offers a reflective framework for understanding contemporary marketing systems, incorporating digital analytics, governance, and social accountability. This integrative approach advances marketing thought and practice in various transformative ways.

1. Moving from Prediction to Reflexivity: Viewing marketing systems as self-learning entities that adapt based on their own data, thereby closing the cause-and-effect loop.

2. Moving from Optimization to Accountability: Incorporating ethical considerations and governance into marketing analytics, acknowledging that algorithms serve moral and managerial roles.

3. Moving from Firm-Centric Control to Co-Evolution: Considering markets as collaborative adaptive networks where firms, consumers, and institutions co-create value and meaning through data feedback.

These advancements build on traditional marketing theory by making ethical learning and system adaptation core to marketing intelligence.

They also demonstrate how an Empirics-First approach—focused on observation rather than assumption—can yield conceptual frameworks that are both sound in theory and impactful in practice.

The Empirics-First Path to Reflexive Digital Twin Marketing Theory

To reinforce the study's contributions to theory and methodology, Figure 6 depicts the Empirics-First Pathway to Reflexive Digital Twin Marketing (RDTM) Theory. The diagram shows how real-world marketing phenomena develop through data-driven investigation, reflexive feedback, and conceptual abstraction, leading to new, widely applicable marketing theories. The model views Empirics-First (EF) research not as an alternative but as a necessary step or partner to theory-driven research, highlighting that strong theory arises from engaging with data-rich, dynamic, and ethically complex marketing environments. This recursive process reflects the co-evolutionary logic of the Digital Twin, where ongoing learning connects empirical insights, stakeholder feedback, and theoretical development innovation.

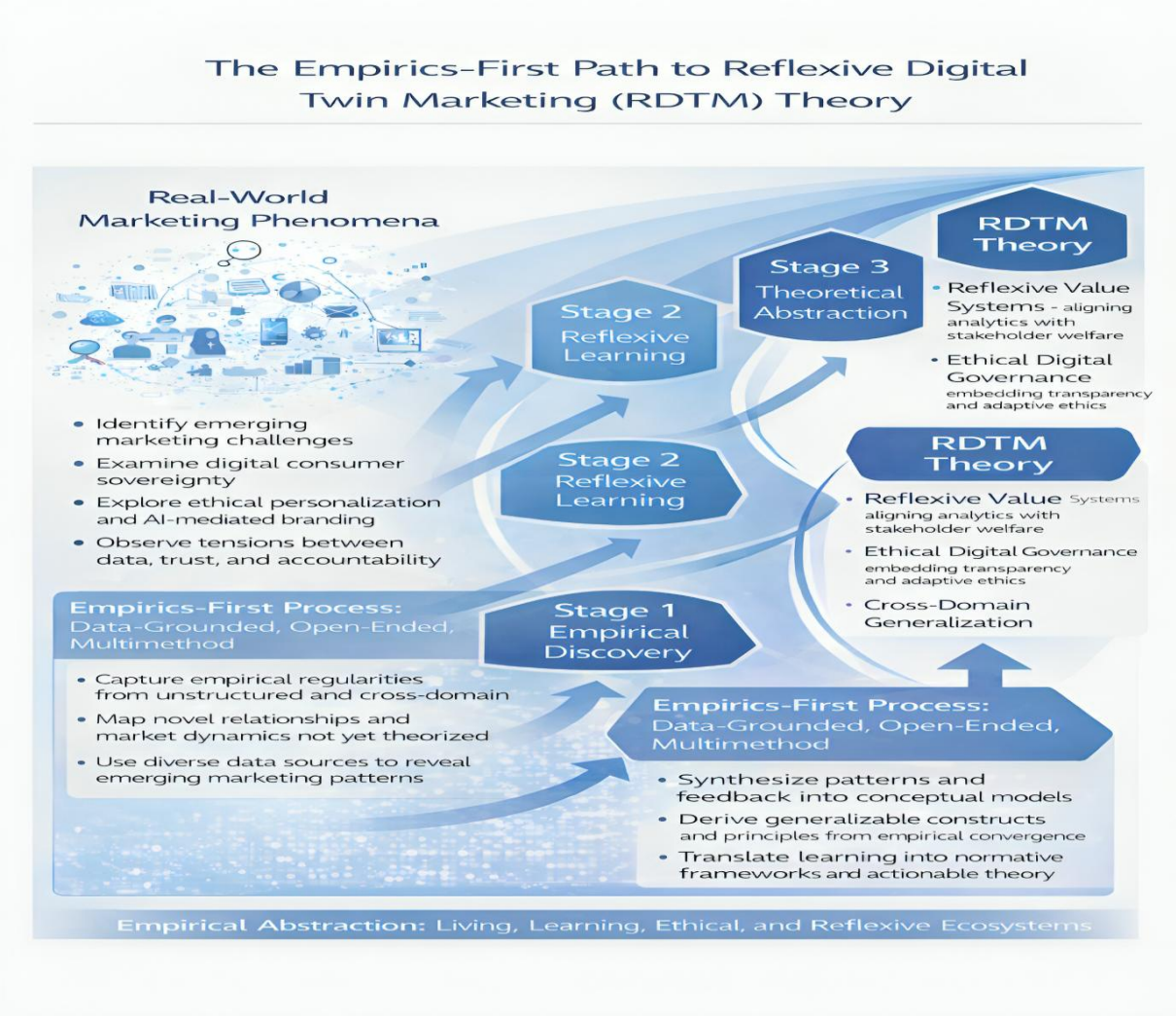

Fig 6: the Empirics-First Approach to Reflexive Digital Twin Marketing (RDTM) Theory — demonstrating the recursive connection among real-world phenomena, data-driven insights, and theoretical development within the Reflexive Evaluation Framework (REF).

Practical Implications

The RDTM framework provides practical pathways for managers, policymakers, and marketing system designers to responsibly leverage AI and data.

• Strategic Simulation: Marketers can utilize Digital Twin environments to experiment with strategies prior to implementation, predicting both commercial and ethical results.

• Reflexive Dashboards: Combining ethical, environmental, and social indicators with traditional KPIs fosters flexible, transparent decision-making.

• Dynamic Governance: Companies can establish participatory feedback mechanisms where customers and stakeholders jointly influence data policies, fostering trust and legitimacy.

These approaches redefine marketing success, shifting focus from mere transactions to adaptable value creation that balances profit with purpose.
For society, such reflexive marketing practices improve transparency, protect digital autonomy, and promote the co-evolution of technology and trust.

Limitations

While this study offers an integrated and reflective synthesis connecting Digital Twin (DT) systems, marketing analytics, and ethical governance, several limitations should be recognized. First, its empirical basis relies on secondary data from existing literature. Although PRISMA-ScR and VOS analyses ensured comprehensive and transparent coverage, the lack of primary empirical data restricts testing causal mechanisms within the emerging Reflexive Digital Twin Marketing (RDTM) Theory. Second, the conceptual synthesis highlights theoretical consistency across healthcare, marketing, and governance, but these fields differ in data maturity and technological adoption. As a result, reflexive mechanisms such as adaptive governance and feedback elasticity may function differently across industries, data infrastructures, or regulatory frameworks. Third, the timeframe (2010–2025) reflects Digital Twin's early adoption phase, meaning new developments in AI and algorithmic governance beyond this period could alter or extend the proposed theory. Lastly, as an Empirics-First (EF) approach, this study favours inductive, data-driven discovery over hypothesis testing.[43] This strengthens contextual validity but may limit generalizability. Future EF research should

triangulate multiple datasets and methods—connecting conceptual insights with practical measurement and longitudinal analysis.

Future Research Directions: The Empirics-First Pathway

Aligned with the Empirics-First philosophy, future research should prioritize real-world phenomena that challenge current theories and create opportunities for inductive insights. Three especially promising avenues are:

1. Empirical Discovery: Investigate unexplored marketing areas—such as consumer digital twins, algorithmic empathy, and AI-driven loyalty programs—to uncover new behavioral patterns.

2. Reflexive Experimentation: Use longitudinal and experimental methods to see how feedback loops affect marketing adaptability and stakeholder trust.

3. Cross-Domain Generalization: Apply RDTM principles to fields like sustainability, policy communication, and social innovation to evaluate their transdisciplinary relevance. By anchoring theory in emerging empirical realities, marketing research can stay relevant in a world shaped by intelligent, adaptable systems.

Closing Reflection: Toward Reflexive Marketing Science

This research reimagines the future of marketing as a reflexive science—an evolving discipline that develops alongside the intelligent systems it creates and studies.
The Reflexive Digital Twin Marketing Theory presents both a framework and a perspective: a model that demonstrates how marketing systems learn and adapt, and a mindset that emphasizes responsibility, transparency, and co-evolution as key to gaining a competitive edge. By embracing an Empirics-First approach, marketing scholars and practitioners can connect data with meaning and insight while maintaining ethical standards. In today's era of digital

reflexivity, marketing's greatest potential lies not just in prediction, but in its capacity to understand and respond to the impact of its own intelligence.

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